05_task_invariant_subspace

October 8, 2026

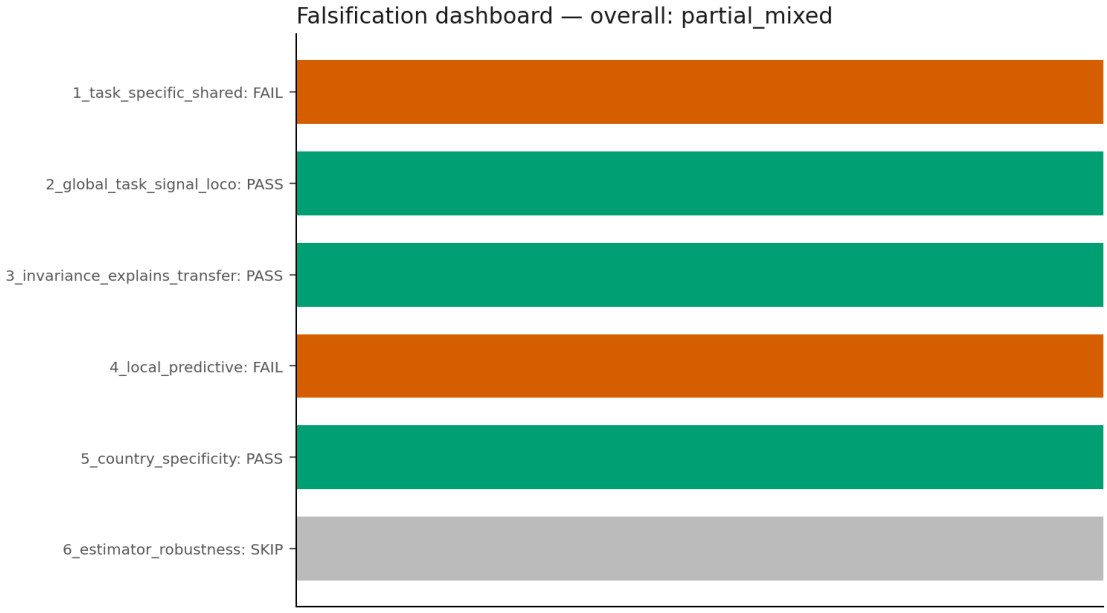
1 Experiment 05 — Task-invariant vs geography-specific subspaces

Run: `task_invariant_subspace_20261006_001647`

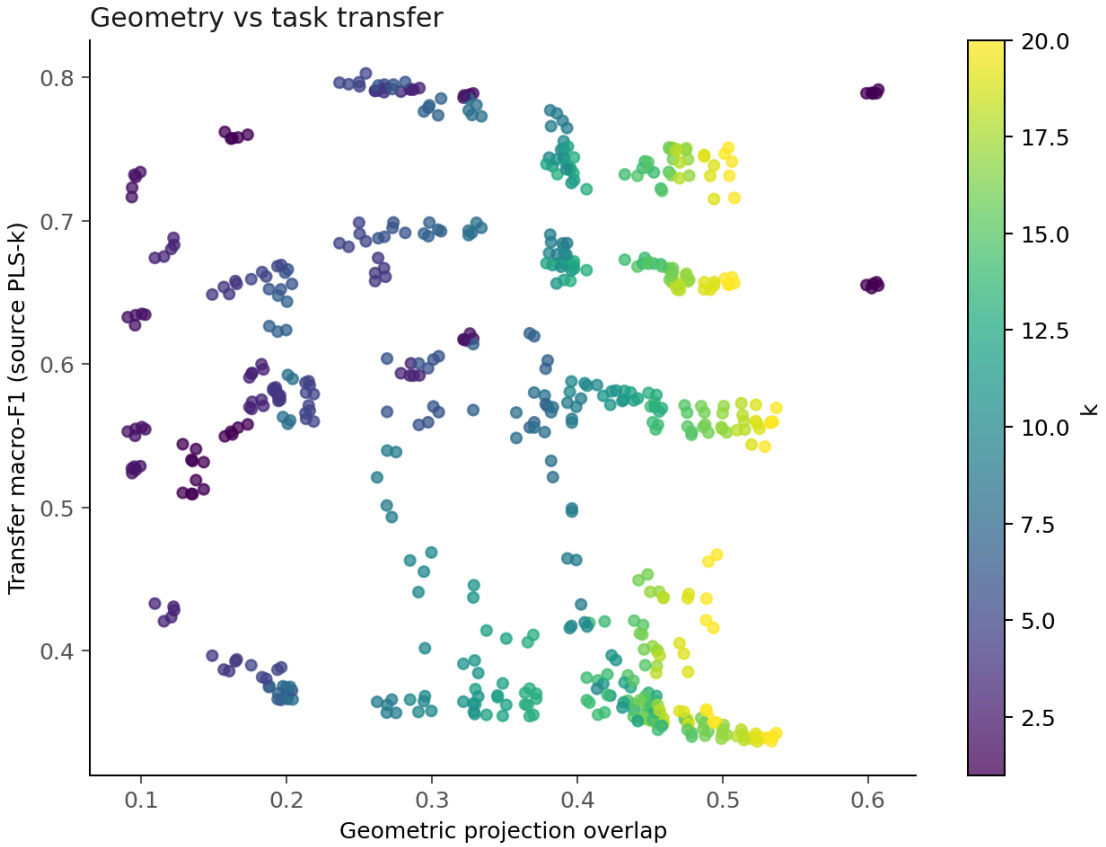
PLS is a **probe** for task-relevant directions. Hypothesis: predictive structure decomposes as $S_c \approx S_G^{\text{task}} + S_{L,c}$ where S_G is defined by **cross-country predictive invariance**, not Grassmann consensus alone (Exp04 spectrum failed vs shuffle).

Figures below load from `../runs/task_invariant_subspace_20261006_001647/figures/` — no recompute needed.

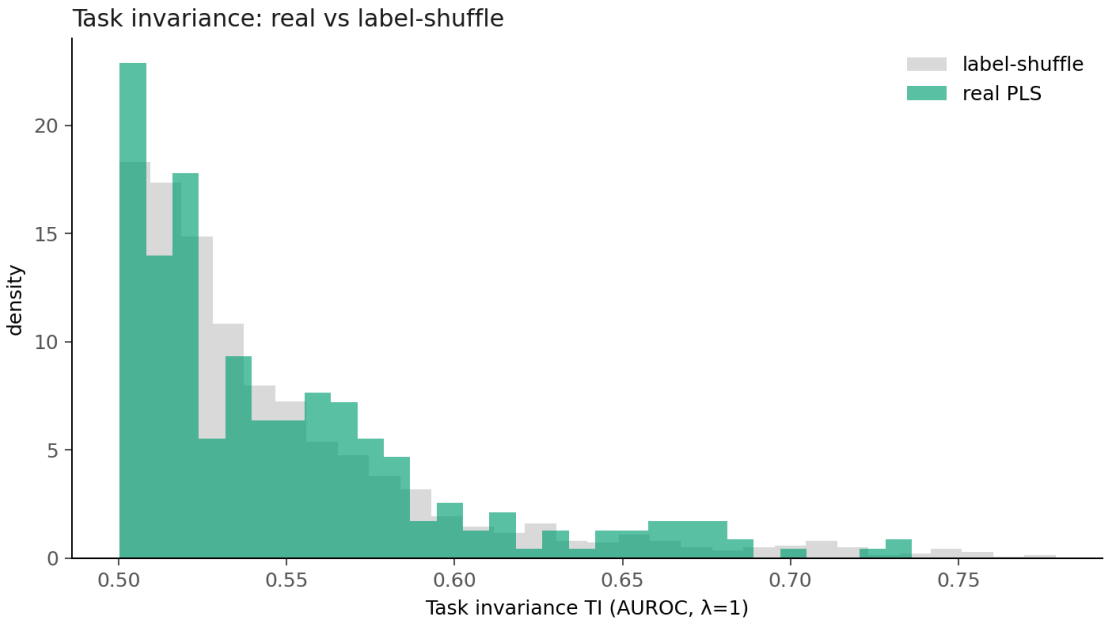
1.1 Falsification dashboard

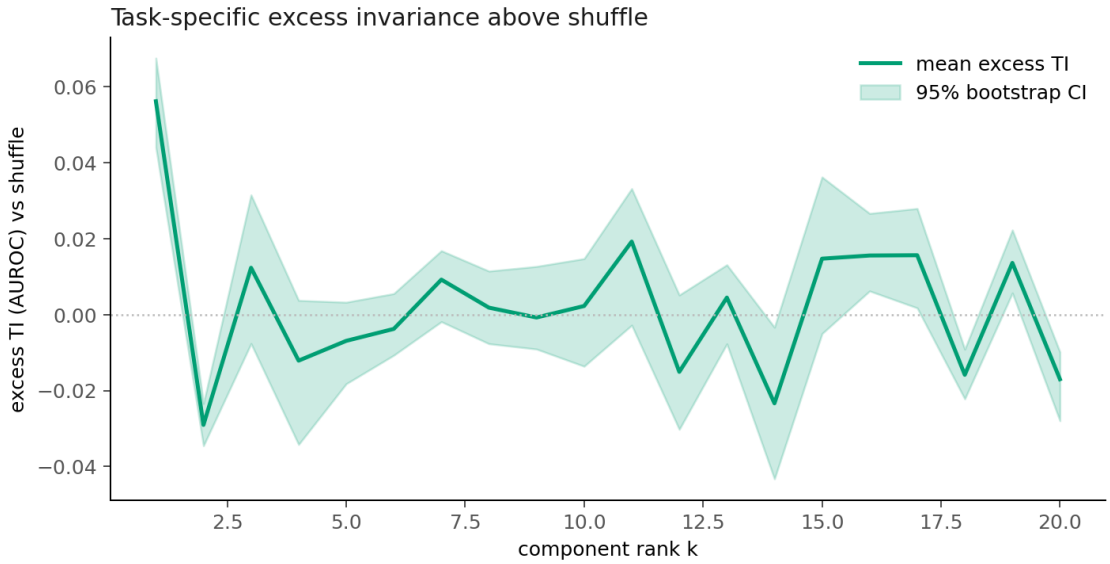


1.2 Geometry vs task

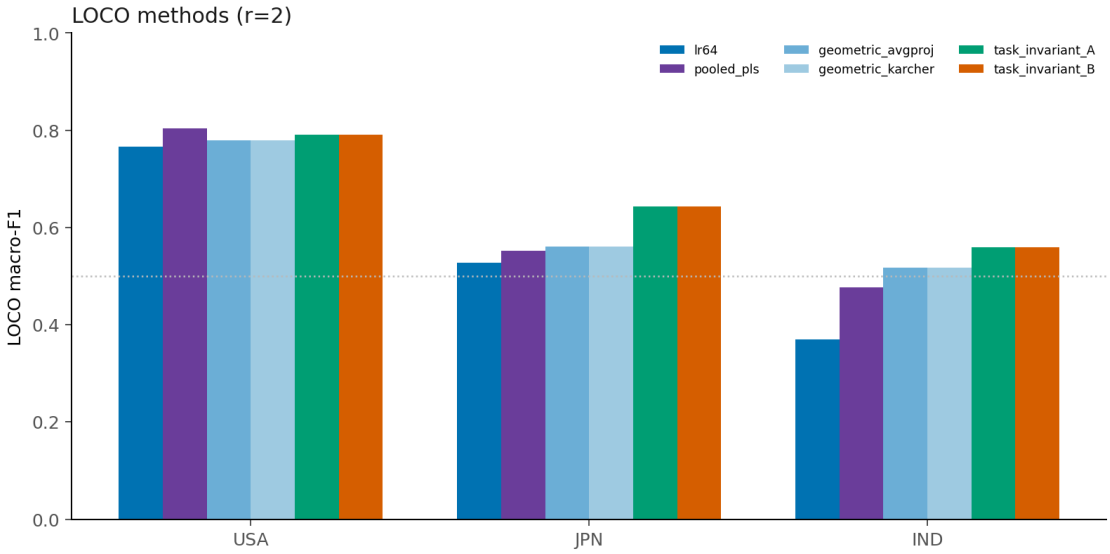


1.3 Task invariance vs shuffle

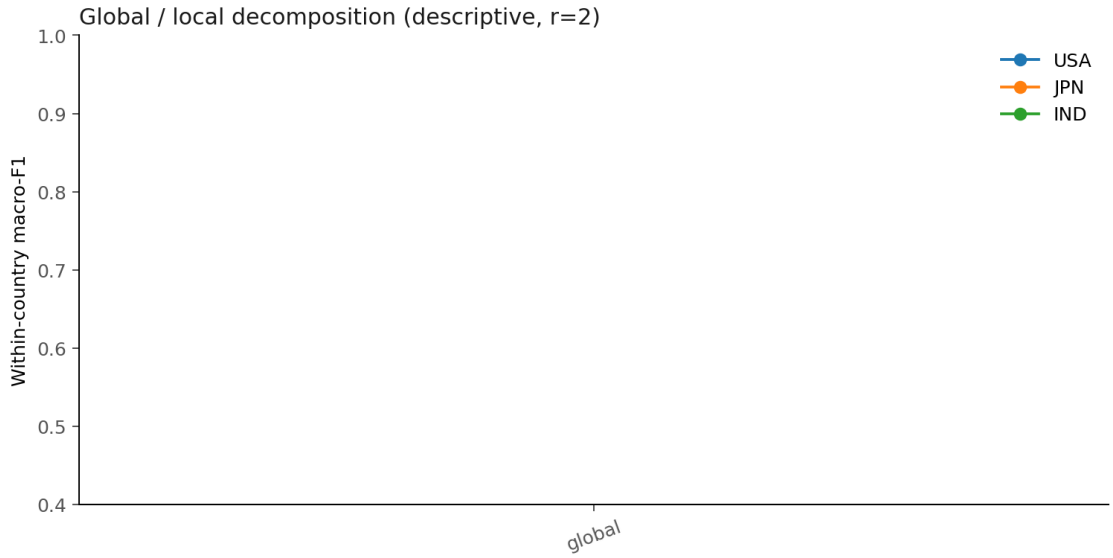




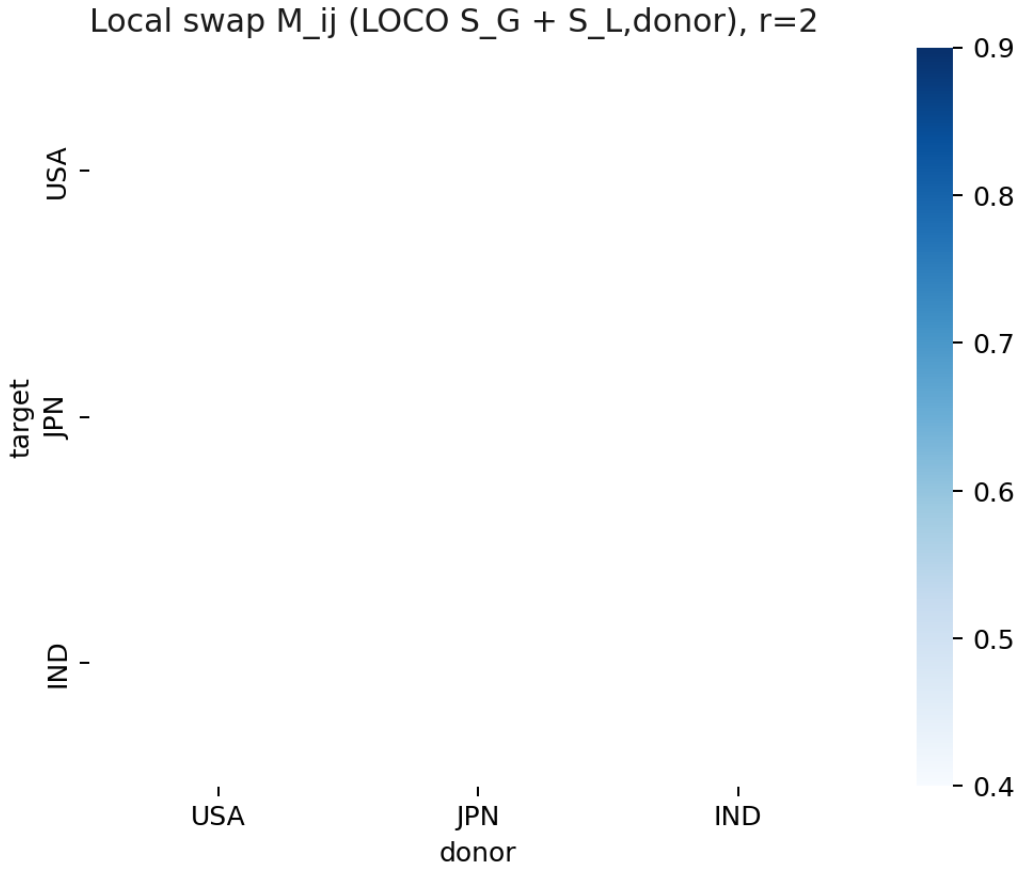
1.4 LOCO methods



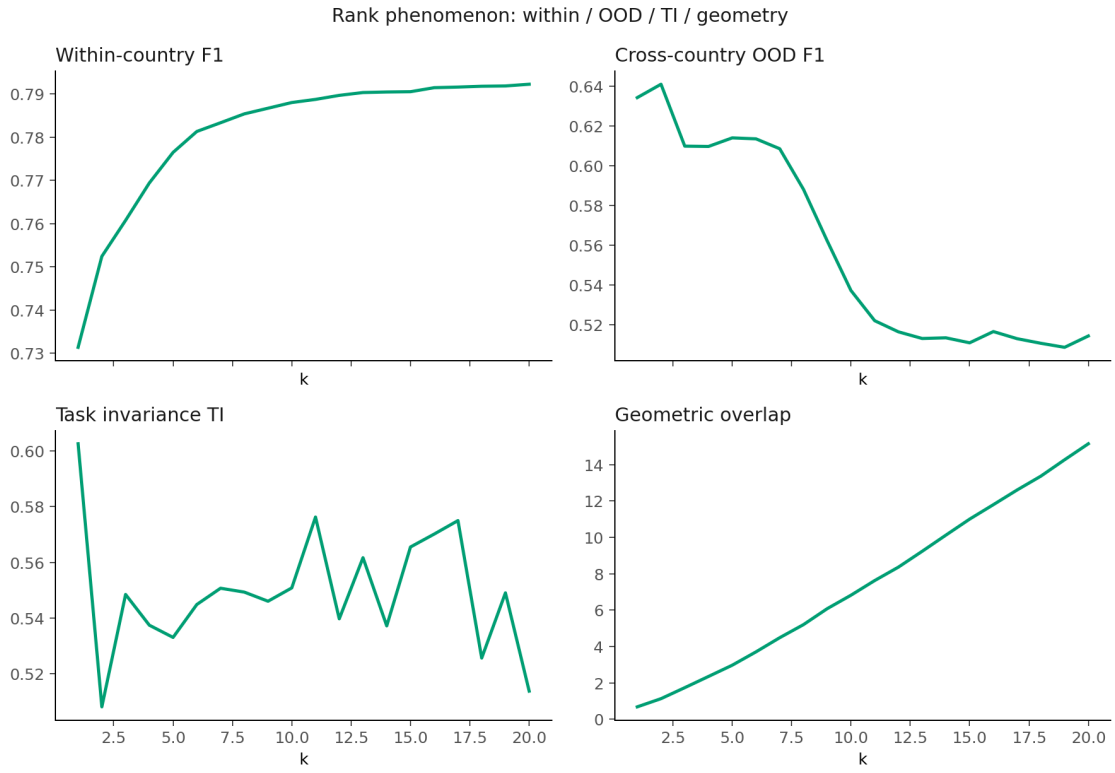
1.5 Global / local



1.6 Local swap



1.7 Rank phenomenon



1.8 Reload artifacts

[ ]:
```
from pathlib import Path
import json, pandas as pd
RUN = Path(r"/home/kv133/Alpha-Earth-Embeddings-Research/week5/runs/
 ↪task_invariant_subspace_20261006_001647")
print((RUN/"REPORT.md").read_text()[:1500])
display(pd.read_csv(RUN/"artifacts"/"loco_results.csv").
 ↪groupby(["r","method"])["macro_f1"].mean().unstack("method").round(3))
print(json.loads((RUN/"artifacts"/"success_criteria.json").
 ↪read_text())["overall"])
```